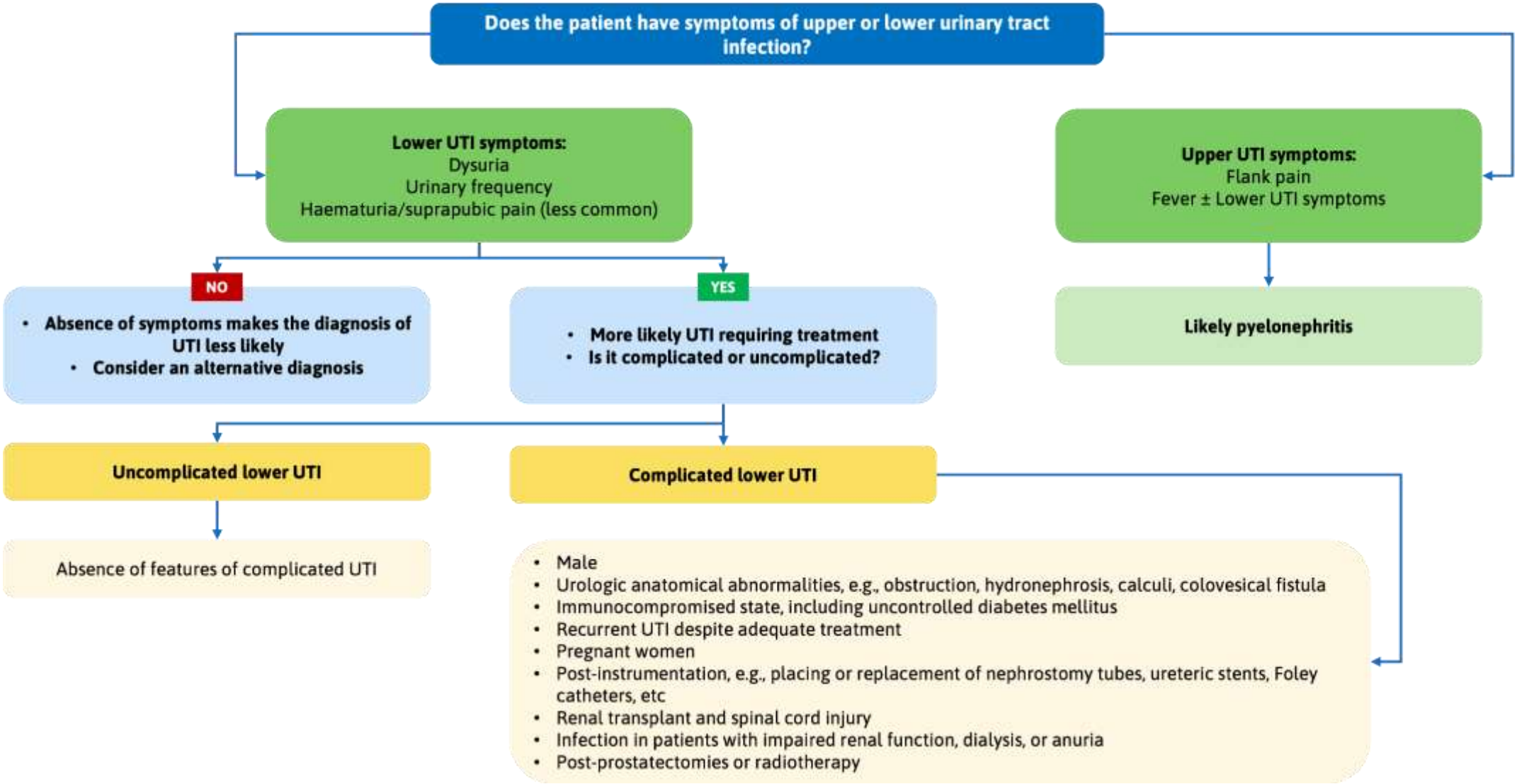
3.9

ADULTS

Urology Infections

Urinary Tract Infection Pathway



Principles of Urinary Tract Infection (UTI) Management

1. Obtain a thorough history including sexual history.
2. Elderly patients may present with atypical symptoms such as confusion and hypothermia.
3. Do rectal and genital examination to exclude prostatitis, posterior urethritis, epididymitis in all men suspected of UTI.
4. Urine culture should only be sent if the patient has symptoms of UTI, screening in pregnancy and prior to certain urological procedures.
5. Culture should be interpreted in parallel with severity of signs and symptoms, as false negatives can occur.
6. Do not treat asymptomatic bacteriuria unless pregnant or undergoing a urological procedure/manipulation, as it does not reduce mortality or morbidity.
7. Unnecessary antibiotic use leads to complications such as *Clostridioides difficile* infection, emergence of multidrug resistance and other adverse drug effects.

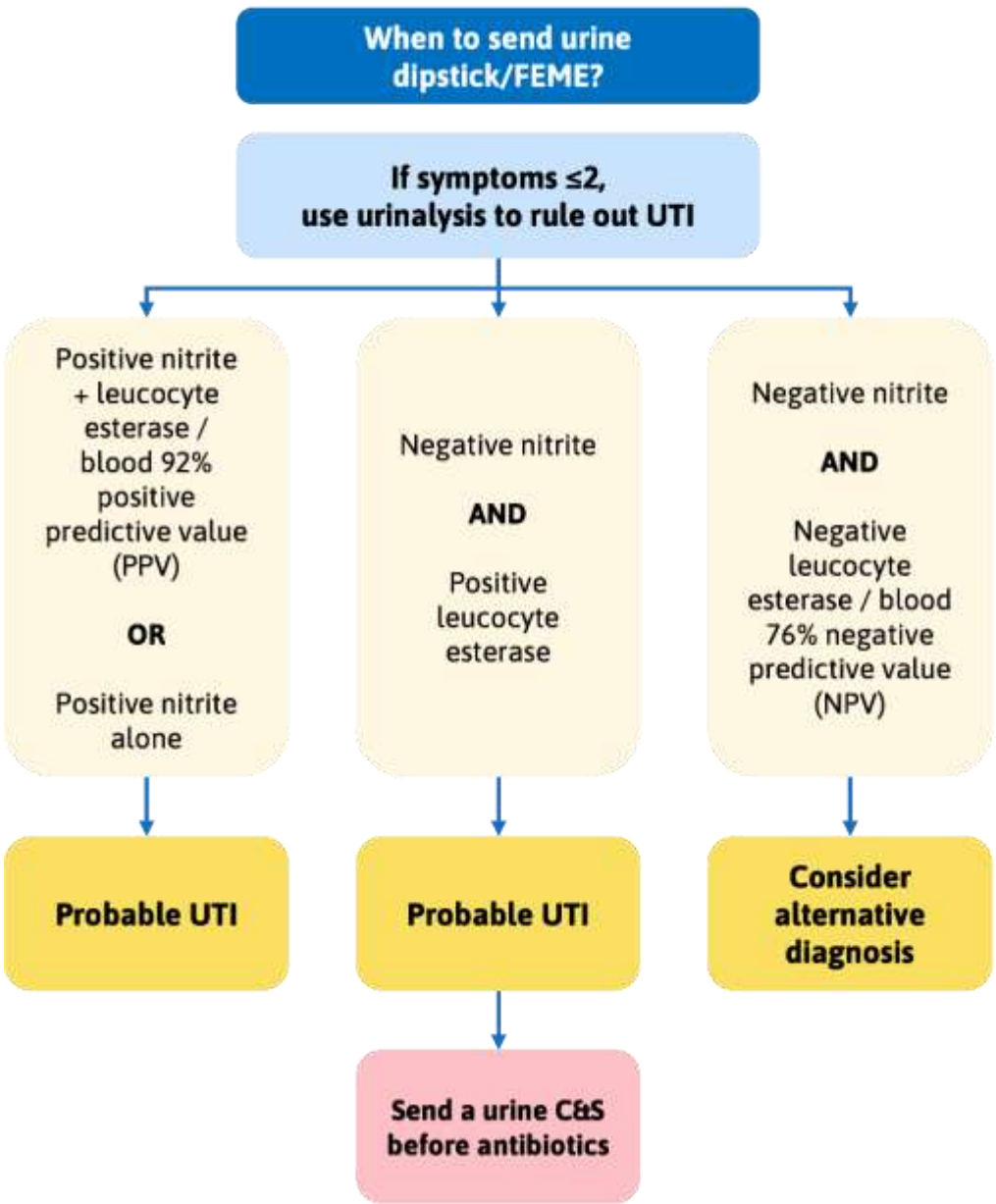
When to send a urine culture?

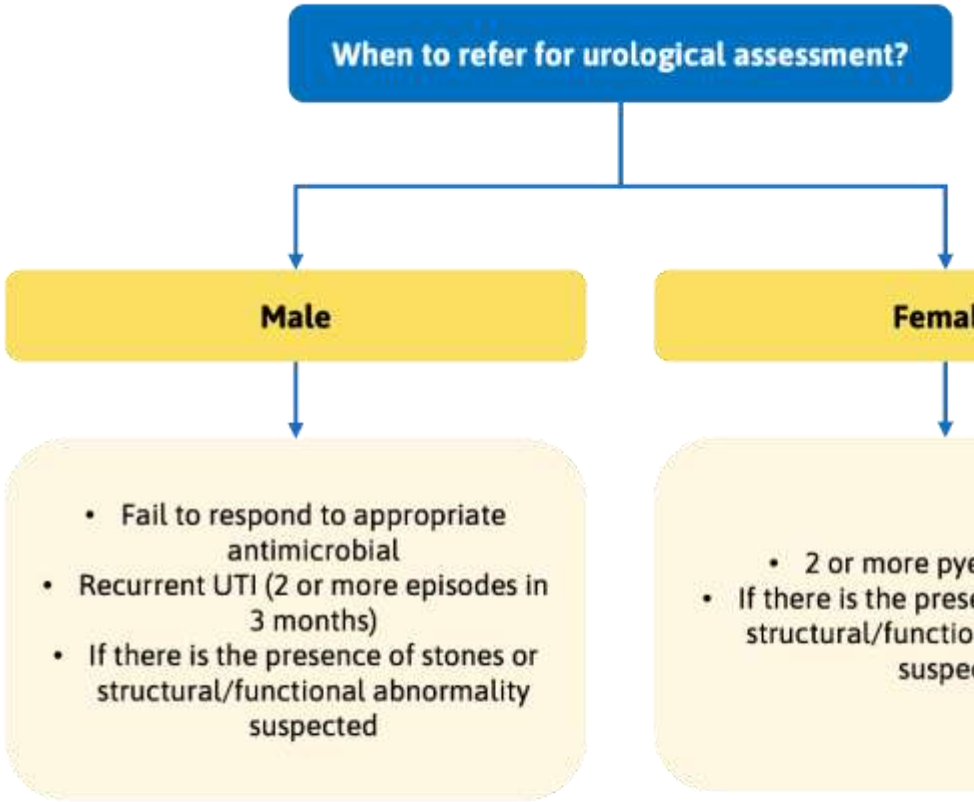
1. Suspected recurrent UTI (>2 in 6 months, ≥3 in 12 months)
2. From aged-care facility with symptoms suggestive of UTI **OR** non-specific symptoms suggestive of an infection
3. Women with suspected complicated UTI/upper UTI (pyelonephritis) or sepsis/chills and rigor
4. Women with symptoms that do not resolve or recur within 2 to 4 weeks of treatment
5. Pregnant women for antenatal screening
6. All men with symptoms of UTI
7. Suspected UTI with hospitalisation for more than 7 days in the last 6 months
8. Previous antibiotic-resistant UTI

When to perform imaging?

1. Clinically ill
2. Severe pyelonephritis
3. Suspected renal abscess
4. Suspected infected renal cysts
5. Recurrent symptoms within a few weeks of treatment
6. Suspected urinary tract obstruction
7. Persistent clinical symptoms despite 48 to 72 hours of appropriate antimicrobial therapy

Urinary Tract Infection Pathway





CHAPTER 3.9	UROLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Urinary Tract Infections			
Asymptomatic bacteriuria Urine bacterial growth ≥105 cfu/mL in 2 serial samples in people without UTI symptoms	Screening for, and treating asymptomatic bacteriuria is not recommended, except: • In pregnant women, **OR** • Prior to transurethral resection of prostate (TURP) or urological procedures breaching the mucosa. Whenever indicated, treatment should be guided by urine culture and sensitivity.		• Duration of treatment in pregnant women: 2-7 days • Duration of treatment in urological procedures: ◦ Targeted antibiotic administration should begin on the night or immediately before procedure ◦ Short course (1 or 2 doses) is recommended for simple procedure e.g., for percutaneous nephrolithotomy (PNL), single dose preoperatively is adequate • For invasive procedure: until catheter removal or maximum 3-5 days post-operative.
Acute cystitis; uncomplicated Symptomatic infection in non-pregnant women with structurally and functionally normal	Nitrofurantoin* 50-100 mg PO q6H for 5 days **OR** Cefalexin 500 mg PO q6H for 5 days	Co-amoxiclav 625 mg PO q8H for 5 days **OR** Co-trimoxazole** (480 mg) 2 tablets PO q12H for 3 days	• *Avoid Nitrofurantoin if GFR < 30 mL/min. • **Co-trimoxazole is not recommended in pregnancy. • ***Consider Fosfomycin for patients with suspected MDR gram-negative infection.

CHAPTER 3.9	UROLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
urinary tract		**OR** Fosfomycin*** 3 g PO single dose	• Fluoroquinolones are not recommended as empirical treatment in UTIs due to: • Selection resistance • Potential serious adverse events i.e., aortic aneurysm or dissection, tendinopathy or tendon rupture and peripheral neuropathy
Acute cystitis; complicated UTI Symptomatic infection in men OR pregnant women OR a urinary tract in which there is a structural or functional abnormality	**Oral Therapy:** Nitrofurantoin* 100 mg PO q6H for 7 days **OR** Cefalexin 500 mg PO q6H for 7 days	**Oral Therapy:** Co-amoxiclav 625 mg PO q8H for 5-7 days **OR** Co-trimoxazole**(480 mg) 2 tablets PO q12H for 7 days **OR** Fosfomycin*** 3 g q48H PO for 3-5 doses	• Acute complicated cystitis: • Male • History of childhood UTI • Immunocompromised • Pregnant • Underlying metabolic disorder (i.e., uncontrolled DM) • Urologic abnormality (i.e., stones, stents, indwelling catheter, neurogenic bladder, polycystic kidney disease) • Symptoms > 7 days • Recent urinary tract manipulation • IV to oral conversion when condition improves.
	Parenteral Therapy: Co-amoxiclav 1.2 g IV q8H	**Parenteral Therapy:** Cefuroxime 1.5 g IV q8H	

CHAPTER 3.9	UROLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Catheter associated UTI (CA-UTI) UTI occurring in a person whose urinary tract is currently catheterized or has been catheterized within the past 48 hours. The presence or absence of odorous or cloudy urine alone should not be used to differentiate catheter associated asymptomatic bacteriuria (CA-ABU) from CA-UTI.	**If no upper UTI symptoms present:** Refer to "Acute cystitis; complicated UTI". **If presence of upper UTI symptoms:** Refer to "Pyelonephritis".		• Duration: • 7 days (if prompt resolution) • 10 - 14 days (if slow response) • To change the catheters and do not repeat urine culture unless indicated (have signs and symptoms of UTI). • Consider Piperacillin/Tazobactam or Cefepime if the patient is colonized/ has a history of *Pseudomonas aeruginosa*. • Consider Meropenem if the patient is colonized/ has a history of ESBL infection.
Acute prostatitis with sexually transmitted disease (STD) risk	Ceftriaxone 500 mg IM single dose (or 1 g IM if weight ≥150 kg) + Doxycycline 100 mg PO q12H for 7 days*	Ceftriaxone 500 mg IM single dose + Azithromycin 1 g PO single dose**	• A thorough sexual history needs to be elicited regardless of age. • Test for STD in all (HIV/Hep B/Hep C/Syphilis test/Gonorrhoea). • *Doxycycline course should be extended to 21 days in the presence of bloody discharge, perianal or mucosal ulcers, or tenesmus and a positive rectal chlamydia test.

CHAPTER 3.9	UROLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
			• **Azithromycin 500 mg PO q24H for 21 days for *C trachomatis.* • Metronidazole 500 mg PO q8H for 14 days for *T vaginalis.*
Prostatitis, prostatic abscess	Co-amoxiclav 1.2 g IV q8H **Once improved/outpatient therapy, change to oral:** Co-trimoxazole (480 mg) 2 tablets PO q12H	Co-amoxiclav 875/125 mg PO q8H **OR** Cefalexin 500 mg PO q6H	• Duration: • 10 - 14 days (prostatitis); could be longer depending on clinical response. • Until resolution (abscess) • Ensure cultures are taken before starting antibiotics. • For abscesses, please ensure adequate source control.
Chronic prostatitis Symptoms that persist for at least for 3 months. **Symptoms:** pain at various locations including the perineum, scrotum, penis, and inner part of	Ciprofloxacin 500 mg PO q12H	Co-trimoxazole (480 mg) 2 tablets PO q12H	• Duration: 4 – 6 weeks. • Reassess after 2 weeks of antimicrobial therapy. Only continue antibiotics if pre-treatment cultures are positive and/or symptoms improve. • Consider non-bacterial prostatitis if symptoms not improved.

CHAPTER 3.9	UROLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
the leg as well as LUTS.			• Doxycycline 100 mg PO q12H for 10 days for C. *trachomatis* or *mycoplasma* infections. • Metronidazole 400 mg PO q8H for 14 days for *T. vaginalis* infections and review after 2 weeks of treatment.

Pyelonephritis/Urosepsis

Considerations for hospitalisation in patients with acute pyelonephritis

1. Comorbid conditions (e.g., renal dysfunction, urologic disorders, diabetes mellitus, advanced liver, or cardiac disease)
2. Hemodynamic instability
3. Male sex
4. Metabolic derangement (e.g., renal dysfunction, acidosis)
5. Pregnancy
6. Severe flank or abdominal pain
7. Toxic appearance
8. Unable to take liquids by mouth
9. Very high fever (> 39.4°C)

* Physicians must be alert for the presence of severe sepsis and septic shock, which require urgent specialized management that is beyond the scope of this review.

CHAPTER 3.9	UROLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Pyelonephritis/Urosepsis			
Community-onset; mild Mild systemic symptoms, low grade fever, loin pain, no nausea or vomiting	Co-amoxiclav 625 mg PO q8H	Cefalexin* 500 mg PO q6H	• Duration: 7-10 days • *Preferred choice in pregnancy • Targeted therapy should be based on antimicrobial susceptibility results.
Community-onset; severe Systemic symptoms (fever, nausea, vomiting, severe pain)	Co-amoxiclav 1.2 g IV q6H	Ceftriaxone 2 g IV q24H	• Total duration (IV and PO): 10 - 14 days (not to exceed 21 days). • Take urine and blood cultures in hospitalised patients • *Use Meropenem if there is risk for ESBL/CRE or the patient is in septic shock.
Hospital-onset	Cefepime 2 g IV q8H	Meropenem* 1 g IV q8H	
Haemodialysis Catheter Related Infections			
Exit-site infection	Cloxacillin 500 mg PO q6H **OR** Cloxacillin 1 g IV q6H	Cefalexin 500 mg PO q12H	• Unless there is exudate at the exit site, routine microbiologic culture of the exit site is not useful. • If there are signs or symptoms or concern of systemic infection, blood cultures should also be obtained.

CHAPTER 3.9	UROLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
			• Duration: 7-14 days (depending on severity i.e., superficial/deep and type of organism). • If exit-site /tunnel infection is due to *Pseudomonas*, refer ID.
Tunnel infection	Cefazolin 2 g IV three times weekly, after dialysis **OR** Cefazolin 1 g IV q24H, after dialysis on HD days	Cloxacillin 2 g IV q6H **OR** Vancomycin* 20 mg/kg IV single dose, followed by dosing recommendation by TDM	• *For patients with penicillin allergy. • Cultures from drainage from the tunnel or exit site and blood cultures from the CVC need to be obtained. • The catheter should be removed, without exchange over a wire. A new catheter should be inserted at a separate site. Alternatively, re-tunnelling could be done. • Duration: 10 – 14 days in the absence of concurrent bacteraemia.
Peritoneal Dialysis (PD) Related Infections			
For prophylaxis related patients on peritoneal dialysis, please refer to **Chapter 3.18: Non-surgical prophylaxis.**			
PD-related exit-site infection	Co-amoxiclav 625 mg PO q12H	Cefalexin 500 mg PO q6H	• Duration: 14 days. • If exit-site/tunnel infection is due to *Pseudomonas*, refer ID.
PD peritonitis*	Cefazolin IP	Cloxacillin** IP	• *Criteria: when at least 2 out of 3 of the following are

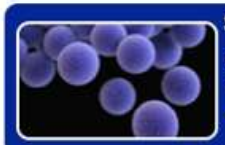
CHAPTER 3.9	UROLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	+ Ceftazidime IP	1 g single dose then 250 mg q6H + Ceftazidime IP **OR** Cefepime IP	present: • clinical features consistent with peritonitis, i.e., abdominal pain and/or cloudy dialysis effluent; • dialysis effluent white cell count > 100/μL or > 0.1 × 10^9/L (after a dwell time of at least 2 hours), with > 50% polymorphonuclear; • positive dialysis effluent culture • Cefazolin is preferred over Cloxacillin for its longer drug stability after preparation. • **Stability of Cloxacillin in peritoneal dialysate is not established. • Duration: 14-21 days depending on organism type. • For fungal prophylaxis, refer to **Chapter 3.18: Non-surgical prophylaxis**

Intraperitoneal Antimicrobial Dosing		
Cefazolin	Intermittent	15 to 20 mg/kg per exchange, once daily
	Continuous	Loading dose of 500 mg/L bag (1 g/2 L bag), maintenance 125 mg for every litre (250 mg/2 L bag q6H)

Ceftazidime	Intermittent	1 to 1.5 grams per exchange, once daily
	Continuous	Loading dose of 500 mg/L bag (1 g/2 L bag), maintenance 125 mg for every litre (250 mg/2 L bag q6H)
Cefepime	Intermittent	1 gram per exchange, once daily
	Continuous	Loading dose of 500 mg/L bag (1 g/2 L bag), maintenance 125 mg for every litre (250 mg/2 L bag q6H)

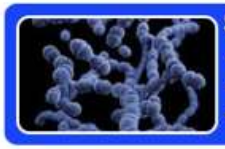
PD Peritonitis

Empirical: IP Cefazolin + IP Ceftazidime



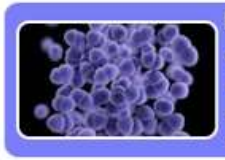
Staphyloccocci (Staph. aureus, Coagulase-negative staphylococci)

- MSSA/ MSCoNS: IP Cefazolin
- MRSA/ MRCoNS: IP Vancomycin
- Duration: 21 days (Staph aureus), 14 days (CoNS)



Streptococci

- Penicillin susceptible: IP Penicillin or IP Cefazolin
- Penicillin resistant: IP Vancomycin
- Duration: 14 days



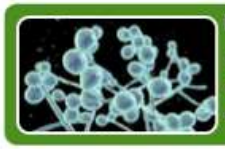
Enterococci

- Penicillin susceptible: PO Amoxicillin
- Penicillin resistant: IP Vancomycin
- VRE: PO/IV Linezolid, refer ID
- Duration: 21 days



Tuberculous peritonitis

- MTB complex: ATT for at least 12 months
- NTM: At least 2 active agents including amikacin for 6 weeks after catheter removal, refer ID



Fungal

- Candida: IV/PO Fluconazole
- Aspergillus: IV Amphotericin B
- Duration: 2-4 weeks after catheter removal

Assess clinical improvement and repeat dialysis effluent cell count and culture at day 3-5.

If no clinical improvement by day 5 (Refractory peritonitis), for catheter removal and continue antibiotics for 14 days



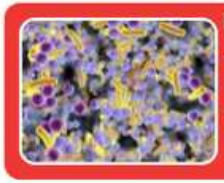
Enterobacterales

- Sensitive strain: IP Ceftazidime
- ESBL: IP Meropenem or PO Ciprofloxacin or PO TMP-SMX
- Amp-C organisms: IP Cefepime or IP Meropenem or PO Ciproflaxin or PO TMP-SMX
- CRE: Refer ID
- Duration: 21 days



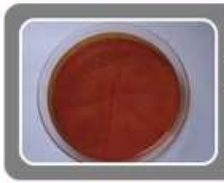
Enviromental gram-negative

- Pseudomonas sp: 2 active antibiotics e.g IP Ceftazidime + PO Ciprofloxacin
- Acinetabacter sp.: IP Ampicillin/sulbactam +/- IP Colistin (if CRAB)
- Stenotrophomonas sp.: PO TMP-SMX + other active antibiotic
- Duration: 21 days



Polymicrobial

- Multiple gram-negative or mixed gram-negative/gram positive: Consider GI problem and obtain surgical assesment; add PO Metronidazole
- Multiple gram-positive: consider touch contamination or PD catheter infection



Culture negative

- If clinically improve: continue with IP Cefazolin for 14 days
- If no clinical improvement: Evaluate for fastidious organisms, mycobacteria, nocardia, filamentous fungus and consider catheter removal

Indication for PD catheter removal:

PD catheter-related peritonitis

Refractory peritonitis. However, if PD effluent WCC is decreasing but does not clear up by day 5, watchful observation without catheter removal may be considered.

Relapsing, recurrent or repeat peritonitis episodes

Pseudomonas peritonitis with concomitant exit-site and tunnel infection

Fungal or NTM PD peritonitis

Dialysis Central Line-Associated Bloodstream Infection (CLABSI)

Catheter MUST be removed in

1. CLABSI presenting with septic shock
2. CLABSI caused by certain pathogens: *S. aureus*, non-fermenting Gram-negative bacilli, *Candida* spp. or Mycobacterium
3. Metastatic complications (endocarditis, thrombophlebitis, or septic pulmonary embolism)
4. Bacteraemia (or candidemia) persisting after 72 hours of adequate treatment
5. Pus is observed at the insertion site
6. Signs of infection at the subcutaneous tunnel
7. No possibility of antibiotic lock therapy

Catheter (cuffed and tunnelled) retention can be considered in

1. Coagulase Negative *Staphylococci* (CoNS) CLABSI (except *S. lugdunensis*)
2. Gram-negative bacilli CLABSI
3. *Enterococca*l CLABSI (Only consider in stable patients without local or systemic complications)
4. Corynebacterium (except *Corynebacterium jeikeium*) CLABSI
5. Suspected CLABSI (i.e., whilst waiting for the blood culture results) provided that the patients are/have:

- hemodynamically stable,
- without immunosuppressive therapy, intravascular foreign bodies, or organ transplantation,
- no suppuration at the insertion site,
- on-going bacteraemia/fungemia

The use of antimicrobial lock therapy (ALT)

1. ALT can be considered as a catheter salvage strategy, and it should be given with systemic antibiotic therapy for CRBSI. The only time that ALT can be used alone is when patients have multiple positive catheter-drawn with concurrent negative peripheral blood cultures involving coagulase-negative *Staphylococci* or Gram-negative bacilli.
2. ALT should be dwelled for not more than 48 hours except for HD patients; the lock solution can be renewed after every dialysis session.

Antibiotics and agents which can be used as ALT		
Vancomycin	Cefazolin	Ceftazidime
Ciprofloxacin	Gentamicin	Ampicillin
Teicoplanin	Daptomycin	Ethanol

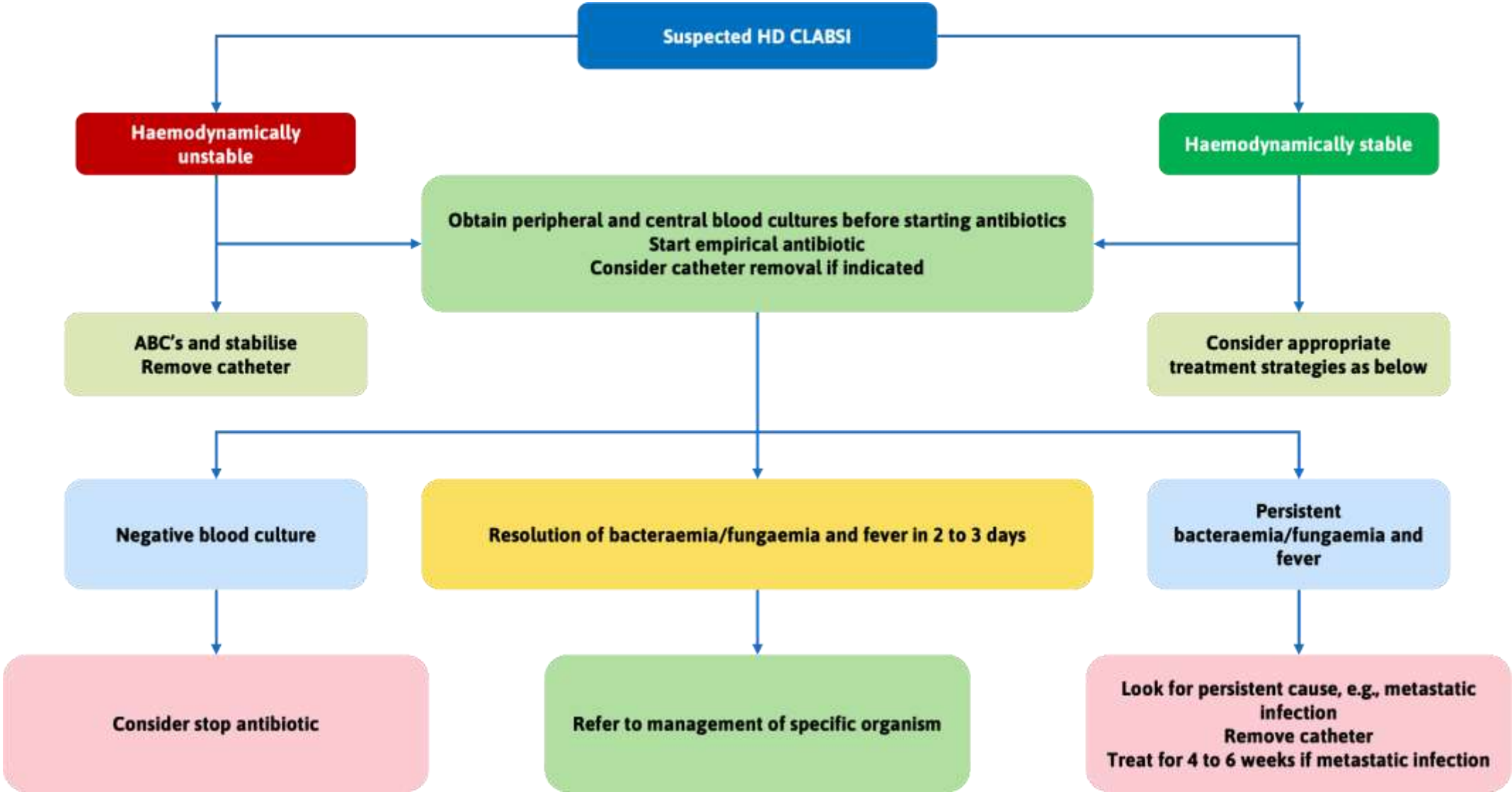
The ideal antibiotics for ALT should have the following characteristics

1. High activity against biofilms (ability to penetrate and disrupt the biofilm)
2. Able to achieve high concentrations (100-1000 times the MIC of planktonic cells)
3. Prolonged stability at room temperature over several days (enables prepared solutions to be stored and the antibiotic lock to be replaced every 24-72 hours)
4. Compatible with anticoagulants
5. Safe
6. Low potential for resistance
7. Cost-effective

Preferred choice of antibiotics for ALT	
Methicillin-susceptible *Staphylococci*	Cefazolin 5 mg/mL

Methicillin-resistant *Staphylococci*	Vancomycin 2 mg/mL (5 mg/mL is more efficacious than 1 mg/mL in eradicating *Staphylococci* embedded within biofilm)
Gram-negative bacteria	Gentamicin 2 mg/mL or Ceftazidime 0.5 mg/mL
Ampicillin-sensitive *Enterococcus* species	Ampicillin 10 mg/mL
Ampicillin-resistant *Enterococci* other than vancomycin-resistant *Enterococci*	Vancomycin 2 mg/mL
Mixed of gram-positive and gram-negative bacterial infections	Ethanol lock 70%

Note: The maximum concentration of ciprofloxacin is limited because of precipitation at higher concentrations.


Suspected HD CLABSI
Haemodynamically unstable
Haemodynamically stable
Obtain peripheral and central blood cultures before starting antibiotics
Start empirical antibiotic
Consider catheter removal if indicated
ABC's and stabilise
Remove catheter
Consider appropriate treatment strategies as below
Negative blood culture
Resolution of bacteraemia/fungaemia and fever in 2 to 3 days
Persistent bacteraemia/fungaemia and fever
Consider stop antibiotic
Refer to management of specific organism
Look for persistent cause, e.g., metastatic infection
Remove catheter
Treat for 4 to 6 weeks if metastatic infection

Coagulase-negative staphylococci	*S. aureus*	Enterococci	Gram-negative bacilli	Candida sp.
1. **MSCoNS:** Cloxacillin OR Cefazolin 2. **MRCoNS:** Vancomycin 3. ECHO is not mandatory 4. If catheter is removed, antibiotic is for 5-7 days. 5. If catheter is retained, antibiotic is for 10-14 days ± ALT 6. *S. lugdunensis* is to be managed as *S. aureus*	1. **MSSA:** Cloxacillin OR Cefazolin 2. **MRSA:** Vancomycin + adjunctive Cefazolin 3. ECHO is mandatory 4. Look for evidence of metastatic infection 5. Remove catheter 6. Repeat blood C&S every 48-72H 7. Antibiotic for 14 days if simple bacteraemia with catheter removal 8. Refer ID	1. Ampicillin or Vancomycin depending on sensitivity 2. ECHO is not mandatory 3. Catheter removal recommended 4. If catheter is removed, antibiotic is for 7-14 days. 5. If catheter is retained, antibiotic is for 14 days + ALT	1. Antibiotic choice is based on sensitivity 2. ECHO is not mandatory 3. Removal of catheter is not mandatory 4. If catheter is removed, antibiotic is for 5-7 days. 5. If catheter is retained, antibiotic is for 10-14 days ± ALT	1. Antifungal choice bas sensitivity 2. ECHO is mandatory 3. Removal of catheter i mandatory 4. Look for evidence of metastatic infection 5. Antifungal for 14 days metastatic infection 6. Refer ID & Ophthalmo

CHAPTER 3.9	UROLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Dialysis Central Line-Associated Bloodstream Infection (CLABSI)			
Non-severe illness	Cefazolin 2 g (if next HD is in 48 hours) **OR** 3 g (if next HD is in 72 hours) IV single dose given after dialysis	Cloxacillin 2 g IV q6H + Ceftazidime 1 g IV q24H given after dialysis on	• *Consider Vancomycin if a catheter is inserted > 3 days or if MRSA is suspected. • Monitor trough Vancomycin level.

CHAPTER 3.9	UROLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Dialysis Central Line-Associated Bloodstream Infection (CLABSI)			
	OR Cefazolin 1 g IV q24H given after dialysis on HD day + Ceftazidime 1 g IV q24H given after dialysis on HD day ± Vancomycin** 20 mg/kg (max 1 g) IV single dose, followed by dosing according to TDM	HD day ± Vancomycin* 20 mg/kg (max 1 g) IV single dose, followed by dosing according to TDM	
Severely ill patients, with any of the following condition (s): 1. Critically ill 2. Severe sepsis 3. Neutropenia	Ceftazidime 1 g IV q24H given after dialysis on HD day **OR** Cefepime 1 g IV q24H given after dialysis on HD day +	Piperacillin/Tazobactam 2.25 g IV q8H **OR** Meropenem* 500 mg IV q24H given after dialysis on HD days + Vancomycin* 20 mg/kg	• *Use Meropenem if ESBL is suspected or the patient is in septic shock.

CHAPTER 3.9	UROLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Dialysis Central Line-Associated Bloodstream Infection (CLABSI)			
	Vancomycin* 20 mg/kg (max 1 g) IV single dose, followed by dosing according to TDM	(max 1 g) IV single dose, followed by dosing according to TDM	
Suspected catheter related candidemia	Fluconazole 100-400 mg IV/PO q24H given after dialysis on HD days	Micafungin 100 mg IV q24H	• De-escalate therapy based on antifungal susceptibility results and remove the catheter if the fungal infection is proven. • Risk factors for catheter related candidemia: • Prolonged use of broad-spectrum antibiotics • Colonization due to *Candida* species at multiple sites • Increase Micafungin dose to 150-200 mg IV q24H if the patient is on CRRT as Micafungin adsorbs to the dialysis membrane.

Kidney Transplantation

For prophylaxis related to patients undergoing kidney transplantation, refer to **Chapter 3.17: Surgical prophylaxis**.

Vaccination

1. Ensure up-to-date standard vaccinations at pre-transplant period, include MMR, DTP, Polio, Hib, Hepatitis B.
2. To assess the immunity for the following disease: Pneumococcal, Meningococcal, Influenza, Varicella, Hepatitis A.
3. If they are not immune to these, they should be vaccinated against them. Assessment of immunity can be done based on vaccination history. Additionally, antibody levels may be used to assess the patients' immunity against *Varicella zoster* and Hepatitis A.

Refer to **Guidelines for Adult Immunisation, 3rd Edition** for the standard vaccination.

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